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1 #include <stdio.h>
2 #include <fcntl.h>
3 #include <unistd.h>
4
5 int main()
6 {
7     int a, b;
8     char buffer[100];
9     int n;
10
11     a = open("source.txt", O_RDONLY);
12     b = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
13
14     n = read(a, buffer, 100);
15     write(b, buffer, n);
16
17     close(a);
18     close(b);
19
20     printf("File copied successfully\n");
21
22     return 0;
23 }

```

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yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ gedit q2.c
** (gedit:5462): WARNING **: 08:59:22.628: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:5462): WARNING **: 08:59:22.629: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ echo "Hello, this is a Linux system call program." > source.txt
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ gcc q2.c -o q2
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ ./q2
File copied successfully
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ cat copy.txt
Hello, this is a Linux system call program.
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ sudo apt install strace
[sudo: authenticate] Password:
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 111
  Download size: 737 kB
  Space needed: 2660 kB / 1023 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 0s (2471 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 48265 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ strace cat source.txt
execve("/usr/bin/cat", ["cat", "source.txt"], 0x7fcb5b6deb8 /* 25 vars */) = 0
brk(NULL)                               = 0x6233c7e05000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x79aa698ae000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)

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openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-... 1024) = 1024
read(3, "e 0 0\ndepts /dev/pts devpts rw,... 1024) = 1024
read(3, ",access=client,msize=65536,trans"... 1024) = 1024
read(3, "des=1024,mode=700,noswap 0 0\ntp"... 1024) = 461
read(3, "", 1024) = 0
close(2) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x79aa69445cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "62339df2c000-62339e01e000 r--p 0"... 1024) = 1024
read(3, "0 08:30 14176 "... 1024) = 1024
read(3, "00 08:30 14388 "... 1024) = 1024
read(3, " /usr/lib/x86_64-linux"... 1024) = 1024
read(3, "0 13931 /us"... 1024) = 566
close(3) = 0
sched_getaffinity(7611, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x79aa698ab000
mprotect(0x79aa698ab000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x79aa698ac000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 7611
rt_sigaction(SIGSEGV, {sa_handler=0x62339e2510c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x79aa69445cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x62339e2510c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x79aa69445cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7ffc9d7df358, 0x200, 0, 0) = 4088
statx(AT_FDCWD, /usr/share/coreutils/locales/cat, AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, /usr/share/coreutils/locales/uucore, AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
stat(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...})

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= 0
read(3, "# Common strings shared across a"... 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, /usr/share/coreutils/locales/cat, AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, /usr/share/coreutils/locales/cat/en-US.ftl, O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x79aa69445cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x79aa69445cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x79aa69445cb0}, {sa_handler=0x62339e2510c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x79aa69445cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x79aa69445cb0}, {sa_handler=0x62339e2510c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x79aa69445cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOF|ECHOK|TEXTEN|ECHOCTL|ECHOWE, ...}) = 0
brk(0x6233c7e47000) = 0x6233c7e47000
statx(AT_FDCWD, "source.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_BASIC_STATS|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0777, stx_size=44, ...}) = 0
openat(AT_FDCWD, "source.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0777, st_size=44, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 44
splice(4, NULL, 1, NULL, 44, 0Hello, this is a Linux system call program.
) = 44
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x79aa698ab000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd year/OS/2520030525_Practical$ strace ./q2
execve("./q2", ["/q2"], 0x7ffc4fbb69c0 /* 25 vars */) = 0
brk(NULL) = 0x59fae73f7000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x790b2d5ef000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=28565, ...}) = 0

```

[illegible]